

AI Developer Learning Roadmap

A complete line-by-line curriculum covering fundamentals, machine learning, deep learning, and cloud engineering.

Phase 1: Foundational Skills & Mathematics

PART 1

- **Programming Fundamentals:** Variables, data types, and memory management. Control structures (conditionals, for/while loops) and modular script creation.
- **Object-Oriented Programming:** Classes, objects, inheritance, and polymorphism to build robust code.
- **Data Manipulation & Analysis:**
 - **NumPy:** Multi-dimensional arrays, vectorization, and matrix operations.
 - **Pandas:** DataFrames, data cleaning (handling missing values), filtering, and merging datasets.
 - **Matplotlib & Seaborn:** Data visualization and exploratory data analysis (EDA).
- **Mathematics for AI:**
 - **Linear Algebra:** Vectors, dot products, matrix multiplication, and eigenvalues.
 - **Calculus:** Derivatives, partial derivatives, the chain rule, and gradients.
 - **Probability and Statistics:** Probability distributions, conditional probability, hypothesis testing, and the deviation method for correlation.

Phase 2: Classical Machine Learning

PART 2

- **Supervised Learning:**
 - Regression models: Linear, multiple linear, and polynomial regression.
 - Classification models: Logistic regression, K-Nearest Neighbors (KNN), and Support Vector Machines (SVM).
 - Tree-based models: Decision trees, Random Forests, and Gradient Boosting (XGBoost).
- **Unsupervised Learning:**
 - Clustering algorithms: K-Means and hierarchical clustering.
 - Dimensionality reduction: Principal Component Analysis (PCA).
- **Model Evaluation & Optimization:**
 - Performance metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
 - Techniques: Preventing overfitting/underfitting, cross-validation, Grid Search, and Random Search.

Phase 3: Deep Learning & Neural Networks

PART 3

- **Neural Network Fundamentals:** The perceptron, activation functions (ReLU, Sigmoid, Softmax), forward propagation, and backpropagation.
- **Optimization & Training:** Loss functions (MSE, Cross-Entropy) and optimization algorithms (Gradient Descent, Adam).
- **Regularization:** Techniques including dropout and batch normalization to ensure model generalization.
- **Advanced Architectures:** Convolutional Neural Networks (CNNs) for computer vision and Recurrent Neural Networks (RNNs/LSTMs) for sequential data.
- **Transfer Learning:** Applying pre-trained architectures to specialized datasets.

Phase 4: Modern AI Engineering & Generative AI

PART 4

- **Prompt Engineering & APIs:** Interacting with foundation models (Gemini, Claude, OpenAI) using few-shot prompting, chain-of-thought, and system/user role separation.
- **Retrieval-Augmented Generation (RAG):** Document chunking, embeddings, vector databases (Pinecone, Chroma), and contextual synthesis.
- **Frameworks & Agents:** Orchestrating LLMs using LangChain and LlamaIndex. Tool-calling capabilities and building multi-step agents using LangGraph.

Phase 5: Production & Cloud Deployment

PART 5

- **Infrastructure & Cloud Integration:** Cloud fundamentals using AWS (EC2 for scalable computing, S3 for model and data storage, and DynamoDB).
- **Model Deployment:** Containerization with Docker and serving models via REST APIs using FastAPI.
- **MLOps Basics:** Monitoring model performance, data drift detection, and logging.